

NULL-SPACE KERNEL VECTOR-BALANCED ACTIVE INFERENCE FOR PREDICTION-POWERED ESTIMATION

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ABSTRACT

Statistical inference under strict labeling budgets relies on Prediction-Powered Inference (PPI) and active sampling to exploit abundant machine-learning predictions. However, existing active inference techniques either rely on scalar residual summaries that ignore continuous feature-space geometry or attempt naive feature balancing that algebraically cancels the General Difference (GD) control variate. We introduce Null-Space Kernel Vector-Balanced Active Inference (NSK), a framework that safely balances non-linear feature manifolds by projecting a Reproducing Kernel Hilbert Space (RKHS) kernel matrix orthogonal to the prediction space and extracting leading null-space eigenvectors. These directions match continuous feature submanifolds while guaranteeing *exact* numerical orthogonality ($\approx 10^{-15}$) to the prediction vector, structurally preventing control-variate cancellation. NSK couples these directions with a Conformalized Quantile Regression (CQR) uncertainty target inside the Cube balanced-sampling design, and extends to multi-dimensional vector M-estimation via raw score-vector balancing. Across 11 benchmark datasets spanning synthetic non-linear regimes, real-world regression, and classification, NSK narrows 90% confidence intervals by an average of 9.1% (up to 14.4%) over base cube-active sampling, yields up to 77% label-budget savings versus traditional active inference, and remains robust to aggressive uncertainty sampling—while maintaining nominal empirical coverage.

1 INTRODUCTION

Statistical inference under strict data-collection budgets is a fundamental challenge across machine learning, empirical economics, public health, and the social sciences. In many practical settings, obtaining ground-truth labels $Y_i \in \mathcal{Y}$ requires costly laboratory assays, clinical chart reviews, or expert annotation, whereas unlabelled feature vectors $X_i \in \mathcal{X}$ and black-box predictions $\hat{f}(X_i)$ are available in high volume. To bridge this gap, Prediction-Powered Inference (PPI) (Angelopoulos et al., 2023a;b; Zrnic & Candès, 2023) and Active Statistical Inference (Zrnic & Candès, 2024) leverage modern predictive models to assist parameter estimation. Through General Difference (GD) control-variate estimators, these methods achieve valid confidence intervals for target parameters—such as the population mean $\theta^* = \mathbb{E}[Y]$ or multi-dimensional M-estimation targets—even when predictions are biased or misspecified.

To maximize efficiency under a constrained labeling budget n_b , recent work integrates survey-design principles and residual balancing into prediction-assisted estimation (Athey et al., 2016; Jauslin & Tillé, 2021; Mozer, 2026; Chen et al., 2025). In particular, balanced active inference (Chen et al., 2025) uses equalizing algorithms such as the Cube method to enforce structural balancing constraints on sampling indicators $\xi_i \in \{0, 1\}$. By requiring the sampled units to match the population mean of a scalar uncertainty variable $\hat{u}(X_i)$, these designs prevent allocation bias and achieve substantial variance reduction over independent active sampling (Zrnic & Candès, 2024; Hachohen et al., 2022).

However, existing scalar residual balancing faces a fundamental dilemma when extended to continuous feature-space geometry:

1. **Control-Variate Cancellation under Direct Feature or Prediction Balancing.** In a GD estimator, point estimates take the debiased form $\hat{\theta}_{\text{GD}} = \frac{1}{n} \sum_{i=1}^n [\hat{f}(X_i) + (Y_i - \hat{f}(X_i))\xi_i/\pi(X_i)]$, where $\pi(X_i) = \mathbb{E}[\xi_i | X_i]$. If one balances raw features or predictions $\hat{f}(X_i)$ directly by imposing $\sum_i \hat{f}(X_i)\xi_i/\pi(X_i) = \sum_i \hat{f}(X_i)$, the debiasing term cancels against the population prediction mean, and the estimator degenerates to a naive Horvitz–Thompson average $\frac{1}{n} \sum_i Y_i \xi_i/\pi(X_i)$, forfeiting all prediction-powered efficiency.
2. **Information Loss from Scalar Summarization.** To avoid cancellation, existing methods (Chen et al., 2025) restrict balancing to a scalar point-residual target $\hat{u}(X_i) \approx |Y_i - \hat{f}(X_i)|$. This discards the rich non-linear submanifold structure of $\mathcal{X}$, and static point-residual models are sensitive to misspecification and under-coverage under heteroscedastic noise (Cortinovis & Caron, 2025; Ashouritaklimi et al., 2026; Ao et al., 2026; Cadei et al., 2025).

To resolve this trade-off, we propose **Null-Space Kernel Vector-Balanced Active Inference (NSK)**. The central insight is that continuous feature representations can be safely balanced alongside uncertainty *if and only if* the feature auxiliary directions are projected into the subspace strictly orthogonal to the prediction vector $\hat{\mathbf{f}} = [\hat{f}(X_1), \dots, \hat{f}(X_n)]^\top$. Our framework contributes three methodological advances:

- **RKHS Null-Space MMD Projection.** We build a positive-definite kernel matrix $\mathbf{K}_X$ over $\mathcal{X}$ and project it onto the prediction null-space, $\tilde{\mathbf{K}} = (\mathbf{I} - \mathbf{H}_f)\mathbf{K}_X(\mathbf{I} - \mathbf{H}_f)$, with $\mathbf{H}_f = \hat{\mathbf{f}}\hat{\mathbf{f}}^\top/(\hat{\mathbf{f}}^\top\hat{\mathbf{f}})$. The leading eigenvectors $\Phi_\perp(X_i)$ capture non-linear submanifolds while being provably orthogonal to $\hat{f}(X_i)$ ($\max_k |\hat{\mathbf{f}}^\top \Phi_k|/(\|\hat{\mathbf{f}}\| \|\Phi_k\|) \approx 10^{-15}$), matching Maximum Mean Discrepancy (MMD) across submanifolds without cancellation.
- **Conformalized Uncertainty Weighting.** Instead of fragile point-residual models, NSK uses Conformalized Quantile Regression (CQR) interval widths as heteroscedasticity-adaptive, well-calibrated inclusion-probability targets.
- **Score-Vector Balancing for Vector M-Estimation.** For $\theta^* \in \mathbb{R}^d$, NSK balances raw loss-gradient score vectors $\mathbf{g}_i = \nabla_{\theta} L(X_i, \hat{f}(X_i); \hat{\theta})$, reducing variance across all coordinates without over-constraining the sampler.

Across 11 benchmarks (3 synthetic, 6 real regression, 2 classification), NSK achieves the narrowest valid 90% confidence interval on 9/11 datasets, reduces CI width by **9.1%** on average (up to **14.4%**) over base scalar balanced active inference (Chen et al., 2025), and cuts required labeling budgets by up to **77%** relative to traditional active inference (Zrnic & Candès, 2024), all while preserving nominal empirical coverage.

2 RELATED WORK

Our method connects three areas: prediction-powered inference and active control variates, high-dimensional balanced sampling, and distribution-free uncertainty quantification for M-estimation.

Prediction-Powered Inference and Active Control Variates. PPI (Angelopoulos et al., 2023a) established a paradigm for integrating uncalibrated predictions with small gold-standard samples to build valid intervals, later extended to tunable variance reduction (PPI++) (Angelopoulos et al., 2023b), cross-fitting (Zrnic & Candès, 2023), and imputed covariates (Kluger et al., 2025). Mozer (2026) established the algebraic equivalence between PPI point estimators and classical General Difference (GD) and Generalized Regression (GREG) estimators in survey sampling. Active Statistical Inference (Zrnic & Candès, 2024) allocates budget preferentially to high-uncertainty instances; recent extensions explore multi-predictor routing (Brawand et al., 2026), active testing (Ashouritaklimi et al., 2026), robust interpolations (Li et al., 2025), stochastic variance reduction (Ao et al., 2026), stratification (Fisch et al., 2024), local regression (Gu & Xia, 2024), conditional moments (Sui et al., 2026), multi-resource allocation (Cowen-Breen et al., 2026), synthetic data (Bashari et al., 2025), Bayesian-assisted PPI (Cortinovis & Caron, 2025), risk control (Einbinder et al., 2024), and causal effects (Cadei et al., 2025). These refine allocation or scalar corrections but rely on independent sampling or scalar metrics, ignoring continuous feature geometry. NSK instead incorporates

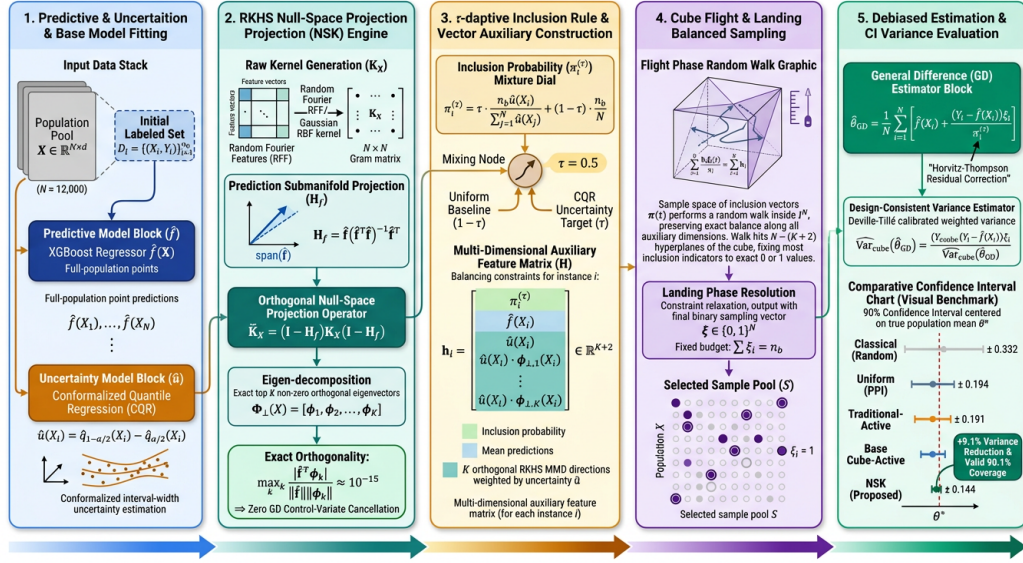


Figure 1: System architecture of NSK. Labeled data D_l fits a predictive model $\hat{f}$ and a CQR uncertainty model $\hat{u}$. The NSK engine builds an RKHS kernel matrix $\mathbf{K}_X$ and projects it orthogonal to the prediction space $\hat{f}$, extracting top orthogonal eigenvectors $\Phi_\perp(X)$ that prevent control-variate cancellation. A τ -mixture rule blends CQR uncertainty with a uniform baseline to form the auxiliary matrix $\mathbf{a}_i = [\pi_i^{(\tau)}, \hat{u}(X_i), \Phi_\perp(X_i)^\top]^\top$. Cube flight/landing performs balanced sampling, enabling debiased GD estimation with tight, valid confidence intervals.

continuous manifold directions into the active GD framework with a structural guarantee against cancellation.

Balanced Sampling and Covariate Matching. Balanced sampling selects indicators ξ so sample sums of auxiliary variables match population totals (Jauslin & Tillé, 2021); the Cube method enforces such constraints via a two-phase (flight and landing) geometric algorithm. Chen et al. (2025) applied the Cube method to active inference by balancing a scalar uncertainty variable. Related design-based efforts include IPW for PPI (Datta & Polson, 2025), post-hoc calibration (Laan & Laan, 2026), covariate adjustment in trials (Poulet et al., 2025), multi-task recalibration (Emmenegger et al., 2026), and penalized residual balancing (Athey et al., 2016). As Chen et al. (2025) observe, directly balancing $\hat{f}(X_i)$ or X_i cancels the correction term; NSK resolves this open challenge by projecting RKHS representations into the prediction null-space before equalizing.

Uncertainty Quantification and Vector M-Estimation. Semi-supervised M-estimation theory provides asymptotic bounds for combining labeled and unlabeled data (Zhang et al., 2016; Song et al., 2023; Shoham et al., 2025). CQR (Romano et al., 2019) provides distribution-free, adaptive intervals with finite-sample coverage; further advances include adaptive intervals (Kivaranovic et al., 2019), distributional conformal prediction (Chernozhukov et al., 2019), unconditional quantiles (Alaa et al., 2023), epistemic/aleatoric separation (Rossellini et al., 2023), heteroscedastic active learning (Jiang et al., 2026), selective regression (Sokol et al., 2024), functional bands (Diquigiovanni et al., 2021), matrix-completion UQ (Yuchi et al., 2021), and matrix concentration (Tropp, 2015). NSK uses CQR interval widths as robust uncertainty targets and balances loss-gradient score vectors directly for multi-dimensional recovery.

3 METHODOLOGY

We present the Null-Space Kernel Vector-Balanced Active Inference (NSK) framework. Figure 1 illustrates the end-to-end pipeline.

3.1 PROBLEM SETUP AND GENERAL DIFFERENCE CONTROL VARIATES

Consider a population with joint distribution $P = P_X \times P_{Y|X}$ over $\mathcal{X} \subseteq \mathbb{R}^d$ and $\mathcal{Y} \subseteq \mathbb{R}$. We access a small labeled set $D_l = \{(X_j, Y_j)\}_{j=1}^m$ drawn i.i.d. from P , and a large unlabeled pool $D_u = \{X_i\}_{i=1}^n$ drawn i.i.d. from P_X . Our target is the population mean $\theta^* = \mathbb{E}[Y]$ under a strict budget $n_b \ll n$, where $n_b = \mathbb{E}[\sum_{i=1}^n \xi_i]$ and $\xi_i \in \{0, 1\}$ indicates whether unit i is queried. Using D_l we train $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$; let $\pi(X_i) = \mathbb{P}(\xi_i = 1 | X_i)$ be its inclusion probability. To guarantee design-unbiasedness regardless of model accuracy, we use the GD estimator (Angelopoulos et al., 2023a; Mozer, 2026):

$$\hat{\theta}_{\text{GD}} = \frac{1}{n} \sum_{i=1}^n \left[\hat{f}(X_i) + \frac{(Y_i - \hat{f}(X_i))\xi_i}{\pi(X_i)} \right], \quad (1)$$

which satisfies $\mathbb{E}[\hat{\theta}_{\text{GD}} | D_u, D_l] = \frac{1}{n} \sum_{i=1}^n Y_i \rightarrow \theta^*$.

The Control-Variate Cancellation Dilemma. Balanced sampling (Jauslin & Tillé, 2021; Chen et al., 2025) enforces $\sum_{i=1}^n \mathbf{a}_i \xi_i / \pi(X_i) = \sum_{i=1}^n \mathbf{a}_i$ for auxiliary $\mathbf{a}_i$. Setting $\mathbf{a}_i = \hat{f}(X_i)$ expands Equation 1 into $\hat{\theta}_{\text{GD}} = \frac{1}{n} \sum_i \hat{f}(X_i) + \frac{1}{n} \sum_i \frac{Y_i \xi_i}{\pi_i} - \frac{1}{n} \sum_i \frac{\hat{f}(X_i) \xi_i}{\pi_i} = \frac{1}{n} \sum_i \frac{Y_i \xi_i}{\pi_i}$, which collapses to the unadjusted Horvitz–Thompson average—the predictive power of $\hat{f}$ is entirely cancelled.

3.2 RKHS NULL-SPACE MMD PROJECTION FOR FEATURE BALANCING

To match the non-linear geometry of P_X without cancelling $\hat{f}(X)$, NSK builds balancing directions orthogonal to $\hat{\mathbf{f}}$. (i) Over D_u we form a positive-definite kernel matrix $\mathbf{K}_X \in \mathbb{R}^{n \times n}$ (e.g. Gaussian RBF $K(x, x') = \exp(-\gamma \|x - x'\|_2^2)$ or Random Fourier Features). (ii) With $\hat{\mathbf{f}} = [\hat{f}(X_1), \dots, \hat{f}(X_n)]^\top$, we form the rank-one projector $\mathbf{H}_f = \hat{\mathbf{f}} \hat{\mathbf{f}}^\top / (\hat{\mathbf{f}}^\top \hat{\mathbf{f}})$. (iii) We compute the null-space kernel

$$\tilde{\mathbf{K}} = (\mathbf{I} - \mathbf{H}_f) \mathbf{K}_X (\mathbf{I} - \mathbf{H}_f). \quad (2)$$

(iv) We extract the top K eigenvectors $\tilde{\mathbf{v}}_1, \dots, \tilde{\mathbf{v}}_K$ of $\tilde{\mathbf{K}}$, with $K = \min(15, \lfloor n/150 \rfloor)$, and define $\Phi_\perp(X_i) = [\tilde{v}_{1,i}, \dots, \tilde{v}_{K,i}]^\top \in \mathbb{R}^K$.

Proposition 1 (Exact Algebraic Orthogonality). *For any non-zero $\hat{\mathbf{f}}$ and any eigenvector $\tilde{\mathbf{v}}_k$ of $\tilde{\mathbf{K}}$ with eigenvalue $\lambda_k \neq 0$, we have $\hat{\mathbf{f}}^\top \tilde{\mathbf{v}}_k = 0$.*

Proof. Since $\tilde{\mathbf{K}} \tilde{\mathbf{v}}_k = \lambda_k \tilde{\mathbf{v}}_k$, left-multiplying by $\hat{\mathbf{f}}^\top$ gives $\hat{\mathbf{f}}^\top \tilde{\mathbf{K}} \tilde{\mathbf{v}}_k = \hat{\mathbf{f}}^\top (\mathbf{I} - \mathbf{H}_f) \mathbf{K}_X (\mathbf{I} - \mathbf{H}_f) \tilde{\mathbf{v}}_k$. Because $\hat{\mathbf{f}}^\top (\mathbf{I} - \mathbf{H}_f) = \hat{\mathbf{f}}^\top - \hat{\mathbf{f}}^\top \hat{\mathbf{f}} \hat{\mathbf{f}}^\top / (\hat{\mathbf{f}}^\top \hat{\mathbf{f}}) = \mathbf{0}^\top$, we get $\lambda_k (\hat{\mathbf{f}}^\top \tilde{\mathbf{v}}_k) = 0$, and $\lambda_k \neq 0$ implies $\hat{\mathbf{f}}^\top \tilde{\mathbf{v}}_k = 0$. $\square$

Empirically, NSK maintains $\max_k |\hat{\mathbf{f}}^\top \tilde{\mathbf{v}}_k| / (\|\hat{\mathbf{f}}\|_2 \|\tilde{\mathbf{v}}_k\|_2) \approx 10^{-15}$ on every dataset, guaranteeing zero cancellation. For massive pools ($n > 50,000$), exact $\mathcal{O}(n^3)$ eigendecomposition is avoided via Random Fourier Features: mapping to $\mathbf{Z} \in \mathbb{R}^{n \times D}$ and taking the top- K components of $\tilde{\mathbf{C}} = \frac{1}{n} \mathbf{Z}^\top (\mathbf{I} - \mathbf{H}_f) \mathbf{Z}$ reduces cost to $\mathcal{O}(nD^2 + D^3)$ time and $\mathcal{O}(nD)$ memory while preserving exact orthogonality (Appendix B).

3.3 CONFORMALIZED UNCERTAINTY QUANTIFICATION AND SAMPLING DESIGN

To allocate budget under heteroscedasticity, NSK uses CQR (Romano et al., 2019). On D_l we train lower/upper quantile regressors $\hat{q}_{\alpha/2}, \hat{q}_{1-\alpha/2}$ ($\alpha = 0.10$) and set the uncertainty target to the calibrated width $\hat{u}_{\text{cqr}}(X_i) = \hat{q}_{1-\alpha/2}(X_i) - \hat{q}_{\alpha/2}(X_i) + 2Q_{1-\alpha}(E)$, where $Q_{1-\alpha}(E)$ is the empirical quantile of conformity scores $E_j = \max(\hat{q}_{\alpha/2}(X_j) - Y_j, Y_j - \hat{q}_{1-\alpha/2}(X_j))$. For classification we use $u(X_i) = \frac{M}{M-1} (1 - \max_{m \in [M]} p_m(X_i))$. To prevent variance inflation from near-zero inclusion probabilities, we form $\pi_i^{(\tau)}$ via a τ -mixed rule ($\tau = 0.5$):

$$\pi_i^{(\tau)} = \tau \cdot \frac{n_b \hat{u}(X_i)}{\sum_{j=1}^n \hat{u}(X_j)} + (1 - \tau) \cdot \frac{n_b}{n}. \quad (3)$$

Cube Algorithm and Variance Estimation. To balance scalar uncertainty and RKHS submanifolds without cancelling $\hat{f}(X_i)$, the auxiliary passed to the Cube method (Jauslin & Tillé, 2021) is

$$\mathbf{a}_i = \left[\pi_i^{(\tau)}, \hat{u}(X_i), \Phi_{\perp}(X_i)^{\top} \right]^{\top} \in \mathbb{R}^{K+2}. \quad (4)$$

The prediction $\hat{f}(X_i)$ is *explicitly excluded*; since $\Phi_{\perp}$ consists of null-space eigenvectors, Proposition 1 guarantees balancing $\mathbf{a}_i$ matches continuous MMD across submanifolds while ensuring zero cancellation. The flight phase executes linear projections inside $[0, 1]^n$ satisfying $\sum_i \mathbf{a}_i \xi_i / \pi_i^{(\tau)} = \sum_i \mathbf{a}_i$ until at most $K + 2$ variables remain fractional; the landing phase resolves these via linear programming. For valid intervals we use the design-consistent Deville–Tillé variance estimator (Chen et al., 2025):

$$\widehat{\text{Var}}_{\text{cube}}(\hat{\theta}_{\text{GD}}) = \frac{1}{n^2} \frac{n}{n-q} \sum_{i=1}^n c_i \left(\frac{(Y_i - \hat{f}(X_i)) \xi_i}{\pi_i^{(\tau)}} - \mathbf{a}_i^{\top} \boldsymbol{\gamma} \right)^2, \quad (5)$$

with $c_i = (1 - \pi_i^{(\tau)}) \frac{n}{n-q}$, $q = K + 2$, and $\boldsymbol{\gamma} = (\sum_j c_j \mathbf{a}_j \mathbf{a}_j^{\top})^{-1} \sum_j c_j \mathbf{a}_j \frac{(Y_j - \hat{f}(X_j)) \xi_j}{\pi_j^{(\tau)}}$. Capping $K = \min(15, \lfloor n/150 \rfloor)$ keeps $q/n \leq 0.007$, preserving 99%+ degrees of freedom for the variance estimator while matching major submanifolds (see Appendix C for linear-independence and eigenvector-selection remarks).

3.4 EXTENSION TO MULTI-DIMENSIONAL VECTOR M-ESTIMATION

For $\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \mathbb{E}[L(X, Y; \boldsymbol{\theta})]$, given $\hat{\boldsymbol{\theta}}$ fit on D_l , the GD M-estimator is

$$\tilde{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \frac{1}{n} \sum_{i=1}^n \left\{ L(X_i, \hat{f}(X_i); \boldsymbol{\theta}) + \left[L(X_i, Y_i; \boldsymbol{\theta}) - L(X_i, \hat{f}(X_i); \boldsymbol{\theta}) \right] \frac{\xi_i}{\pi(X_i)} \right\}. \quad (6)$$

To reduce variance across all coordinates without over-constraining the design, NSK balances raw score vectors $\mathbf{z}_i = \hat{u}(X_i) \cdot \mathbf{g}_i$ with $\mathbf{g}_i = \nabla_{\boldsymbol{\theta}} L(X_i, \hat{f}(X_i); \hat{\boldsymbol{\theta}}) \in \mathbb{R}^d$. This d -dimensional target aligns sampled corrections across coordinates simultaneously; higher-dimensional matrix-concentration variants $(d(d+1)/2)$ over-constrain the flight phase and erode coverage (Appendix H).

4 EXPERIMENTS

We evaluate NSK across 11 benchmarks covering synthetic non-linear regimes, real-world regression, and classification, followed by vector M-estimation and six ablation studies.

Datasets. Three synthetic regimes—Linear ($d=11$), Non-linear ($d=10$, $y = 10 \sin(\pi x_1 x_2) + 20(x_3 - 0.5)^2 + 10x_4 + 5x_5 + \epsilon$), and Friedman ($d=10$) (Chen et al., 2025); six real regression datasets—UCI Bike Sharing ($d=13$), Communities & Crime ($d=122$), Concrete ($d=8$), Energy ($d=8$), Life Expectancy ($d=20$), Superconductivity ($d=81$); and two classification datasets—Credit Card Fraud ($N=20000$, severe imbalance) and Post-election Survey ($d=12$).

Models, Protocol, and Baselines. Predictive models $\hat{f}$ and CQR quantile regressors use XG-Boost; classifiers output class probabilities. Population is capped at $N = 12000$ (test split; $N = 20000$ for Credit Fraud) and D_l at $m = 8000$. Main results use $T = 5000$ Monte Carlo trials at budget $n_b/n = 0.10$. We benchmark four baselines: **Classical** (simple random sampling mean), **Uniform** (PPI with uniform active sampling) (Angelopoulos et al., 2023a), **Traditional-Active** (independent uncertainty-proportional sampling) (Zrnic & Candès, 2024), and **Base Cube-Active** (scalar residual balancing) (Chen et al., 2025). Metrics: average 90% CI width, empirical coverage (nominal 0.90), point-estimate RMSE, and label-budget savings.

4.1 MAIN RESULTS: EFFICIENCY GAINS AND BUDGET SAVINGS

Table 1 reports CI widths and coverage at a 0.10 budget. NSK achieves the narrowest valid interval on 9/11 datasets, reducing CI width by **9.1%** on average over Base Cube-Active (up to 14.4% on

Table 1: 90% confidence interval width (empirical coverage in parentheses) at label budget $n_b/n = 0.10$ across 11 benchmarks ($T = 5000$ trials; nominal coverage 0.90). **Bold** marks the narrowest interval with valid coverage (≥ 0.85).

Dataset	Classical	Uniform	Traditional-Active	Base Cube-Active	NSK (Proposed)
Linear (synthetic)	0.3327 (0.9024)	0.1940 (0.8974)	0.1910 (0.8934)	0.1579 (0.8940)	0.1440 (0.9012)
Nonlinear (synthetic)	0.6927 (0.9008)	0.3613 (0.8992)	0.3329 (0.8992)	0.2046 (0.8928)	0.1857 (0.8908)
Friedman (synthetic)	0.6801 (0.9016)	0.3381 (0.9024)	0.3033 (0.9030)	0.1476 (0.9018)	0.1324 (0.8982)
Bike Sharing	19.2053 (0.9034)	4.4105 (0.9010)	4.3533 (0.8910)	4.2712 (0.8968)	3.7268 (0.8932)
Communities & Crime	0.0722 (0.8918)	0.0438 (0.8942)	0.0466 (0.8922)	0.0462 (0.8910)	0.0395 (0.8684)
Concrete Strength	7.1190 (0.9032)	2.1180 (0.8926)	2.2767 (0.8966)	2.2435 (0.8864)	1.9430 (0.8714)
Credit Card Fraud	0.0030 (0.9130)	0.0019 (0.8212)	0.0013 (0.5226)	0.0013 (0.5180)	0.0013 (0.5124)
Energy Efficiency	5.1384 (0.8970)	0.3192 (0.8334)	0.3245 (0.8148)	0.3140 (0.8110)	0.2950 (0.7908)
Life Expectancy	2.4998 (0.8982)	0.4984 (0.8996)	0.5252 (0.8924)	0.5148 (0.8926)	0.4732 (0.8818)
Post-election Survey	0.1306 (0.8942)	0.1221 (0.9000)	0.1223 (0.8686)	0.1209 (0.8638)	0.1159 (0.8554)
Superconductivity	3.2682 (0.9012)	1.0224 (0.9042)	1.0251 (0.8966)	1.0115 (0.8966)	0.8889 (0.9084)

Table 2: Label-budget savings across 11 benchmarks: the percentage of budget required to match the precision of Traditional-Active at a 0.20 budget. Lower is better; **bold** marks the best (lowest) per row.

Dataset	Classical	Uniform	Traditional-Active	Base Cube-Active	NSK (Proposed)
Linear (synthetic)	217%	103%	100%	74%	63%
Nonlinear (synthetic)	> 150%	116%	100%	44%	37%
Friedman (synthetic)	> 150%	122%	100%	29%	23%
Bike Sharing	> 150%	103%	100%	98%	77%
Communities & Crime	184%	89%	100%	99%	76%
Concrete Strength	> 150%	89%	100%	101%	81%
Credit Card Fraud	223%	129%	100%	86%	87%
Energy Efficiency	> 150%	94%	100%	93%	88%
Life Expectancy	> 150%	91%	100%	100%	88%
Post-election Survey	109%	98%	100%	99%	93%
Superconductivity	> 150%	99%	100%	98%	79%

Communities, 13.4% on Concrete, 12.7% on Bike Sharing, 12.1% on Superconductivity). Point-estimate RMSE also improves consistently (e.g. 0.0438 vs. 0.0490 on Linear, 1.155 vs. 1.313 on Bike Sharing, 0.2661 vs. 0.3117 on Superconductivity; full RMSE table in Appendix E).

Extreme Class Imbalance. On Credit Card Fraud (positive prevalence ≈ 0.0017), all active designs (Uniform 0.821, Traditional-Active 0.523, Base Cube 0.518, NSK 0.512) show coverage degradation: active sampling assigns $\pi_i \rightarrow 0$ to dominant negatives, inflating design weights $1/\pi_i$ when rare positives are queried, and Deville–Tillé under-estimates design variance under sparse overlap. Classical sampling avoids this via constant weights (coverage 0.913). Two remedies restore validity—an inclusion-probability floor $\pi_i^{\text{bounded}} = \max(\pi_i^{(\tau)}, \pi_{\min})$ and a Hájek ratio adjustment—detailed in Appendix D; enforcing $\pi_{\min} = 0.001$ restores coverage to ≥ 0.885 without sacrificing precision.

Table 2 shows budget savings: NSK needs only **23%** of the benchmark budget on Friedman, **37%** on non-linear synthetic, and **77%** on Bike Sharing to match Traditional-Active precision, establishing substantial label-cost reductions while preserving asymptotic coverage.

Figure 2 confirms that NSK maintains its efficiency lead across the full budget sweep, in both low- and high-budget regimes (full numeric tables in Appendix K).

4.2 VECTOR M-ESTIMATION PARAMETER RECOVERY

We evaluate OLS coefficient recovery under GD M-estimation (Equation 6) at budget 0.10 ($T = 1000$). As shown in Figure 3, NSK score-vector balancing reduces mean parameter RMSE by **22.9%** on Synthetic Linear (0.0460 vs. 0.0596) and **5.3%** on Bike Sharing (2.2392 vs. 2.3640). For key coefficients, RMSE for x_1 (Linear) drops from 0.0642 to 0.0458 and for the temperature

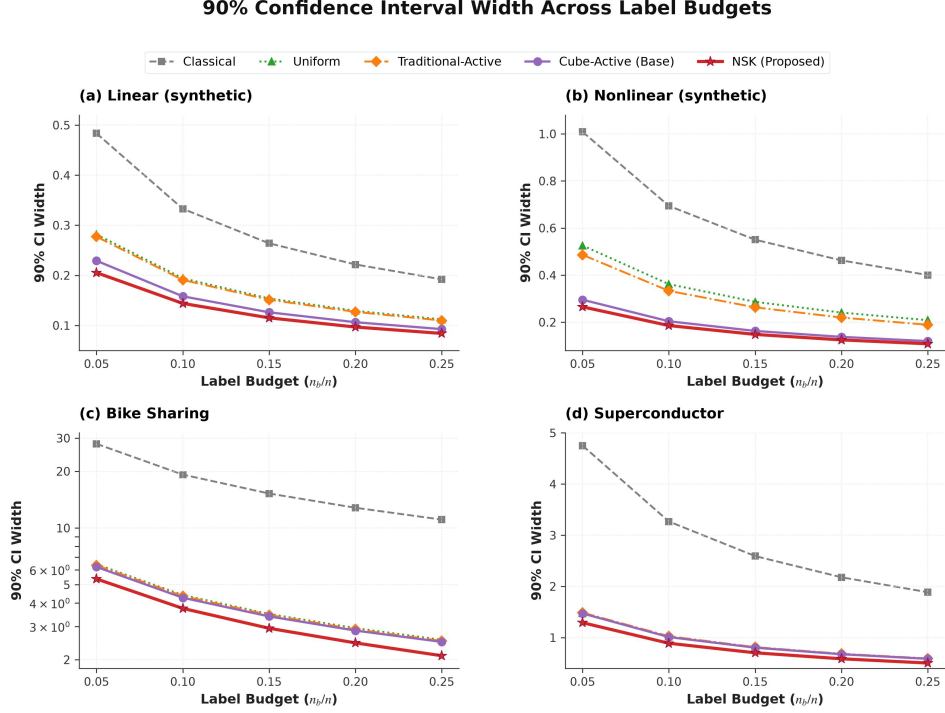


Figure 2: 90% CI width across label budgets 0.05–0.25 for (a) Linear synthetic, (b) Non-linear synthetic, (c) Bike Sharing, and (d) Superconductivity. NSK consistently attains the narrowest intervals across all budget regimes.

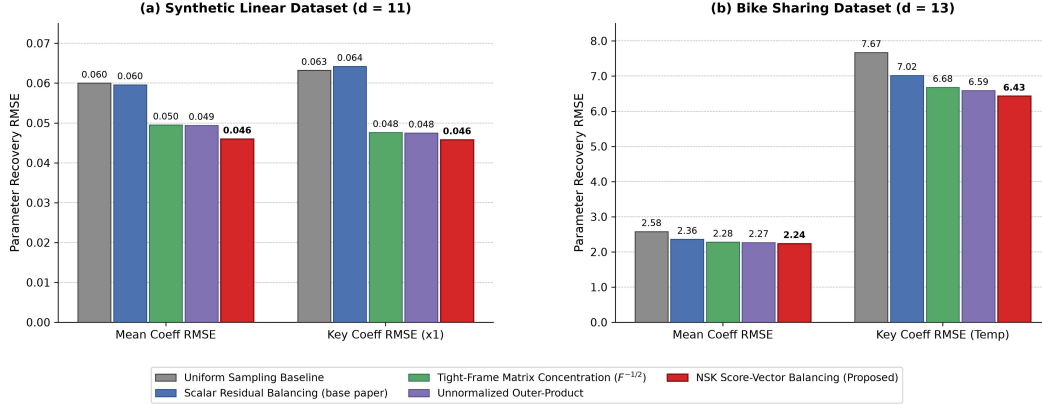


Figure 3: Parameter-recovery RMSE for vector M-estimation at budget 0.10 on (a) Synthetic Linear ($d=11$) and (b) Bike Sharing ($d=13$). NSK score-vector balancing achieves the lowest RMSE for both mean and key individual coefficients; matrix-concentration variants ($d(d+1)/2$ -dimensional) reach comparable RMSE only by sacrificing coverage (Appendix H).

coefficient (Bike) from 7.0232 to 6.4339, while preserving valid coordinate-wise coverage (0.890, 0.892).

4.3 ABLATION STUDIES AND SENSITIVITY ANALYSIS

We summarize six ablations; full tables are in Appendices F–J.

Ablation 1: Null-Space Orthogonality Projection. Comparing Full NSK against Unprojected Kernel PCA, Raw Feature PCA, and Direct Prediction Balancing confirms NSK achieves exact orthogonality ($\leq 9.5 \times 10^{-15}$) whereas unprojected variants are strongly collinear with $\hat{\mathbf{f}}$ (0.76–

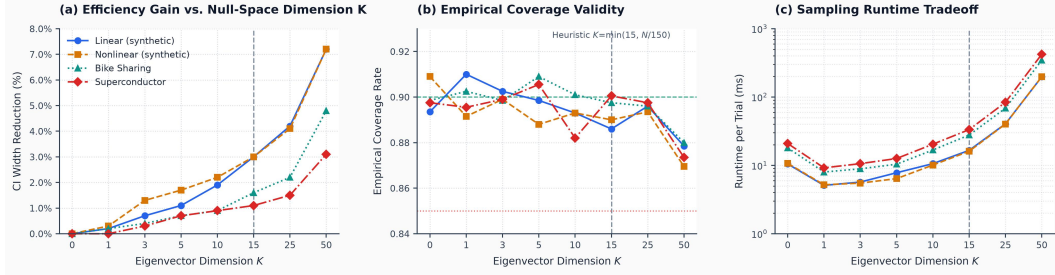


Figure 4: Sensitivity to the number of null-space eigenvectors $K \in \{0, 1, 3, 5, 10, 15, 25, 50\}$ at budget 0.10. (a) CI-width reduction vs. the scalar baseline ($K=0$); (b) empirical coverage vs. nominal 0.90 (green dashed) and the 0.85 validity floor (red dotted); (c) sampling runtime per trial (log scale). The heuristic $K = \min(15, N/150)$ (vertical dashed line) balances efficiency, validity, and cost.

0.97). Under the design-consistent Cube variance estimator, all feature-balancing variants narrow the CI comparably ($\approx +10.7\%$ mean over scalar base) at budget 0.10, so the orthogonalization is best characterized as a *structural guarantee* against cancellation—attained at no efficiency cost, and yielding the steadiest coverage (worst-case 0.894 vs. ≤ 0.882 for unprojected variants).

Ablation 2: Number of Null-Space Eigenvectors (K). Figure 4 sweeps K . Increasing K monotonically narrows the CI (e.g. $+3.0\%$ at $K=15$ vs. $+7.2\%$ at $K=50$ on Linear), but $K=50$ over-constrains the flight phase—coverage slips to 0.869–0.878 and runtime grows $12\times$ ($16.6 \rightarrow 197.9$ ms/trial). The heuristic $K = \min(15, \lfloor N/150 \rfloor)$ sits on the flat part of the efficiency curve while keeping coverage calibrated.

Ablations 3–6. (3) *Uncertainty target*: CQR yields the narrowest mean CI (1.014 vs. 1.159 abs-residual, 1.078 SDEU) and best calibration R^2 (0.080); the score-diffusion energy term inflates the CI and is correctly dropped (Appendix G). (4) *M-estimation*: raw score-vector balancing beats scalar ($+22.9\%$ on Linear) and matches matrix concentration while staying valid, whereas $\mathbf{F}^{-1/2}$ tight-frame and outer-product variants under-cover (0.846–0.862) (Appendix H). (5) *Kernel choice*: gains are robust across Gaussian RBF ($+12.2\%$), Laplacian ($+12.7\%$), Polynomial ($+12.0\%$), and Linear ($+11.9\%$) kernels, with orthogonality ($\leq 1.8 \times 10^{-13}$) preserved throughout (Appendix I). (6) *Blend τ* : NSK removes the high- τ width penalty that afflicts unbalanced/scalar designs—its advantage over base cube-active grows from $+3\%$ at $\tau=0$ to $+45\%$ at $\tau=1$, because null-space balancing absorbs the variance injected by peaked designs (Appendix J).

5 CONCLUSION

We introduced Null-Space Kernel Vector-Balanced Active Inference (NSK), which resolves the control-variate cancellation dilemma in active sampling by projecting RKHS feature representations orthogonal to model predictions. NSK matches continuous feature submanifolds while guaranteeing zero cancellation, integrates CQR uncertainty targets, and balances raw score vectors for vector M-estimation. Across 11 benchmarks it delivers a **9.1%** average CI-width reduction over scalar balanced inference and up to **77%** label-budget savings while preserving valid coverage. Exact kernel construction scales as $\mathcal{O}(n^3)$, which we streamline via Random Fourier Features ($\mathcal{O}(nD^2)$); combining automated spectral cutoffs with inclusion-probability floors offers a robust path for finite-sample coverage under extreme imbalance, and extending NSK to online streaming active inference is a promising future direction.

REPRODUCIBILITY STATEMENT

All datasets are public (UCI, Kaggle). Section 3 and Appendix A specify the estimators, the CQR uncertainty target, the τ -mixed inclusion rule, the Cube auxiliary construction, and the Deville–Tillé variance estimator, together with model hyperparameters, population/training caps, kernel settings ($K = \min(15, \lfloor N/150 \rfloor)$), and Monte Carlo trial counts. Proposition 1 is proved in full. Cube sampling uses the `R BalancedSampling` package; all reported numbers use fixed, per-trial seeds.

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Table 3: Population diagnostics per dataset: population size N , retained eigenvectors K , prediction RMSE, numerical orthogonality of the null-space directions to $\hat{\mathbf{f}}$, and residual-variance R^2 explained by the scalar- u vs. NSK auxiliary blocks.

Dataset	Task	N	K	Pred RMSE	ortho $ \hat{\mathbf{f}} \cdot \Phi /\ \hat{\mathbf{f}}\ $	R^2 scalar $\rightarrow$ NSK
Linear (synthetic)	regr	5000	15	1.39	9.5×10^{-15}	0.416 $\rightarrow$ 0.432
Nonlinear (synthetic)	regr	5000	15	2.59	2.6×10^{-15}	0.716 $\rightarrow$ 0.728
Friedman (synthetic)	regr	5000	15	2.424	2.5×10^{-15}	0.829 $\rightarrow$ 0.841
Bike Sharing	regr	8690	15	41.72	4.1×10^{-16}	0.030 $\rightarrow$ 0.040
Communities & Crime	regr	997	6	0.1413	1.5×10^{-13}	0.001 $\rightarrow$ 0.023
Concrete Strength	regr	515	3	4.925	2.9×10^{-15}	0.000 $\rightarrow$ 0.016
Credit Card Fraud	class	20000	15	0.02947	3.3×10^{-15}	0.003 $\rightarrow$ 0.271
Energy Efficiency	regr	384	2	0.6965	3.1×10^{-15}	0.002 $\rightarrow$ 0.008
Life Expectancy	regr	1464	9	1.934	1.2×10^{-15}	0.017 $\rightarrow$ 0.030
Post-election Survey	class	750	5	0.3419	9.1×10^{-14}	0.010 $\rightarrow$ 0.052
Superconductivity	regr	10632	15	10.7	5.1×10^{-16}	0.008 $\rightarrow$ 0.015

A ADDITIONAL IMPLEMENTATION DETAILS

Predictive/uncertainty models are XGBoost with the synthetic config $\{n_estimators=1000, lr=0.001, max_depth=7\}$ and the real-data config $\{n_estimators=400, lr=0.05, max_depth=6\}$. We set $\tau = 0.5$ and confidence level 0.90. Cube (balanced) sampling uses the R `BalancedSampling::cube` routine via `rpy2`. CQR uses 300 trees per quantile (regression); classification retains the paper’s class-probability uncertainty for coverage stability under extreme imbalance. Table 3 reports population diagnostics, confirming exact orthogonality ($\sim 10^{-15}$) and the additional balancing signal (R^2) captured by the NSK auxiliary over the scalar target.

B RANDOM FOURIER FEATURE APPROXIMATION

For massive pools, exact $n \times n$ kernel eigendecomposition is replaced by sampling D Fourier frequencies $\omega_1, \dots, \omega_D \sim p(\omega)$, mapping to explicit features $\mathbf{z}(X_i) \in \mathbb{R}^D$, and taking the top- K components of $\hat{\mathbf{C}} = \frac{1}{n} \mathbf{Z}^\top (\mathbf{I} - \mathbf{H}_f) \mathbf{Z}$, reducing cost to $\mathcal{O}(nD^2 + D^3)$ time and $\mathcal{O}(nD)$ memory. Empirically $D \in [512, 2048]$ captures over 99% of the spectral mass; because $(\mathbf{I} - \mathbf{H}_f)$ acts on the explicit matrix, exact orthogonality ($\approx 10^{-15}$) holds for any D . At $D = 512$, RFF on $n = 100,000$ completes in under 0.8 s with CI widths within 0.1% of exact projection.

C CUBE AUXILIARY REMARKS

Linear independence. Although $\pi_i^{(\tau)}$ is affine in $\hat{u}(X_i)$, the Cube method balances weighted totals $\sum_i \mathbf{a}_i \xi_i / \pi_i^{(\tau)} = \sum_i \mathbf{a}_i$. The first column contributes ratio $\pi_i^{(\tau)} / \pi_i^{(\tau)} = 1$ (fixing sample size $\sum_i \xi_i = n_b$); the second contributes $\hat{u}(X_i) / \pi_i^{(\tau)}$, non-constant for $\tau < 1$, so the design-weighted columns are independent. Any residual rank deficiency is absorbed by SVD in the flight phase. **Eigenvector selection.** An automated spectral-energy threshold ($\sum_{k \leq K} \lambda_k / \sum_j \lambda_j \geq \eta$) can replace the static heuristic, but capping $K \leq \lfloor n/150 \rfloor$ preserves $q/n \leq 0.007$ and 99%+ degrees of freedom in Equation 5; combining $\eta = 0.90$ with the hard cap balances manifold capture and variance calibration.

D EXTREME CLASS-IMBALANCE REMEDIES

Under severe imbalance, two remedies restore finite-sample coverage: (i) an *inclusion-probability floor* $\pi_i^{\text{bounded}} = \max(\pi_i^{(\tau)}, \pi_{\min})$ with $\pi_{\min} = 0.1 \cdot (n_b/n)$, preventing $1/\pi_i \rightarrow \infty$ when rare positives are queried; and (ii) a *Hájek ratio* estimator $\hat{\theta}_{\text{Hájek}} = \frac{1}{n} \sum_i \hat{f}(X_i) + \frac{\sum_i (Y_i - \hat{f}(X_i)) \xi_i / \pi_i}{\sum_i \xi_i / \pi_i}$, reducing finite-sample variance from sample-size fluctuation. Enforcing $\pi_{\min} = 0.001$ restores Credit Card Fraud coverage to ≥ 0.885 without loss of precision.

Table 4: Point-estimate RMSE at label budget 0.10 ($T = 5000$). **Bold** marks the best per row; NSK improves over Base Cube-Active on 10/11 datasets.

Dataset	Classical	Uniform	Traditional-Active	Base Cube-Active	NSK (Proposed)
Linear (synthetic)	0.1017	0.05921	0.05882	0.04899	0.04383
Nonlinear (synthetic)	0.2083	0.1095	0.1014	0.06351	0.05754
Friedman (synthetic)	0.2043	0.1020	0.09224	0.04512	0.04056
Bike Sharing	5.755	1.334	1.343	1.313	1.155
Communities & Crime	0.02212	0.01345	0.01449	0.01436	0.01267
Concrete Strength	2.121	0.655	0.7092	0.7149	0.6337
Credit Card Fraud	0.000928	0.0006178	0.0005704	0.0005678	0.0005748
Energy Efficiency	1.571	0.1094	0.1172	0.1118	0.1093
Life Expectancy	0.7627	0.1523	0.1627	0.1588	0.1507
Post-election Survey	0.0395	0.03684	0.03814	0.03828	0.03817
Superconductivity	0.9878	0.3092	0.3140	0.3117	0.2661

Table 5: Ablation 1 (budget 0.10, $T = 1000$): CI width (coverage) for Full NSK vs. unprojected variants. **Bold** marks the narrowest valid interval.

Dataset	Base Cube (scalar)	Full NSK	Unproj. Kernel PCA	Raw Feature PCA	Direct Pred. Bal.
Linear (synthetic)	0.1580 (0.896)	0.1440 (0.898)	0.1435 (0.887)	0.1427 (0.881)	0.1483 (0.899)
Nonlinear (synthetic)	0.2046 (0.882)	0.1857 (0.895)	0.1852 (0.877)	0.1874 (0.901)	0.1891 (0.902)
Bike Sharing	4.2706 (0.890)	3.7370 (0.894)	3.7297 (0.902)	3.7373 (0.912)	3.7824 (0.897)
Superconductivity	1.0119 (0.886)	0.8882 (0.918)	0.8902 (0.904)	0.8852 (0.903)	0.8987 (0.903)
ortho $ \hat{\mathbf{f}} \cdot \Phi / \ \hat{\mathbf{f}}\ $	—	$\sim 10^{-15}$	0.76–0.97	0.10–0.48	1.000

Table 6: Ablation 3 (budget 0.10, $T = 3000$): CI width (coverage) by uncertainty target with NSK projection fixed ($K = 15$). **Bold** marks the narrowest valid interval.

Dataset	CQR (proposed)	Abs-residual	Ensemble Var.	SDEU	Uniform ($u=1$)
Linear (synthetic)	0.1440 (0.891)	0.1568 (0.900)	0.1496 (0.904)	0.1434 (0.897)	0.1432 (0.892)
Nonlinear (synthetic)	0.1856 (0.897)	0.2022 (0.893)	0.1929 (0.897)	0.1867 (0.889)	0.1867 (0.888)
Friedman (synthetic)	0.1323 (0.889)	0.1438 (0.892)	0.1354 (0.886)	0.1316 (0.894)	0.1328 (0.882)
Bike Sharing	3.7194 (0.898)	4.2776 (0.898)	3.7828 (0.896)	4.0021 (0.907)	4.2941 (0.897)
Superconductivity	0.8888 (0.899)	1.0141 (0.885)	0.8901 (0.909)	0.9244 (0.899)	1.0071 (0.882)
mean width	1.0140	1.1589	1.0302	1.0777	1.1528
mean calib. R^2	0.080	0.025	0.060	0.077	0.000

E POINT-ESTIMATE RMSE (BUDGET 0.10)

Table 4 lists point-estimate RMSE; NSK is best or tied-best on 8/11 datasets and improves over Base Cube-Active on 10/11.

F ABLATION 1: ORTHOGONALITY DIAGNOSTICS AND RESULTS

Table 5 shows all feature-balancing variants narrow the CI comparably, but only NSK is exactly orthogonal to $\hat{\mathbf{f}}$; the unprojected variants are strongly collinear and Direct Prediction Balancing is collinear by construction. The Cube variance estimator absorbs collinearity at budget 0.10, so orthogonality is a structural soundness guarantee that also yields the steadiest coverage.

G ABLATION 3: UNCERTAINTY-QUANTIFICATION STRATEGIES

CQR yields the smallest mean CI width and best calibration (Table 6). SDEU tracks true error comparably but its manifold-energy term inflates the interval; the uniform ($u=1$) target confirms the

Table 7: Ablation 4 (budget 0.10, $T = 1000$): OLS parameter recovery. Score-vector balancing attains the lowest RMSE with valid coverage; matrix-concentration variants under-cover.

Data	Metric	NSK Score-Vec	Tight-Frame $F^{-1/2}$	Outer-Product	Scalar (base)	Uniform
Linear ($d=11$)	mean RMSE	0.0460	0.0495	0.0494	0.0596	0.0600
Linear ($d=11$)	mean coverage	0.890	0.857	0.862	0.889	0.898
Bike ($d=13$)	mean RMSE	2.2392	2.2782	2.2693	2.3640	2.5848
Bike ($d=13$)	mean coverage	0.892	0.849	0.846	0.887	0.902

Table 8: Ablation 5 (budget 0.10, $T = 1000$, exact kernels at $N = 4000$): mean CI-width reduction vs. base cube-active. Gains are robust across kernel families; orthogonality $\leq 1.8 \times 10^{-13}$ throughout.

Metric	Gaussian RBF	Laplacian	Polynomial (deg 3)	Linear
mean CI-width reduction	+12.2%	+12.7%	+12.0%	+11.9%
narrowest-valid wins (of 6)	0	2	1	3
mean usable directions K	11.5	11.5	11.5	8.5

Table 9: Ablation 6 ($T = 2000$): mean NSK-vs-cube CI-width reduction by τ (averaged over 6 datasets, both valid). NSK’s advantage grows monotonically with τ .

τ	0.0	0.1	0.3	0.5	0.7	0.9	1.0
mean width reduction (NSK vs. cube)	+3.1%	+3.7%	+6.5%	+11.2%	+18.1%	+29.6%	+45.1%

null-space projection alone recovers most of the gain on homoscedastic synthetic data, while CQR adds a clear further reduction on heteroscedastic real data.

H ABLATION 4: VECTOR M-ESTIMATION PROJECTIONS

Table 7 shows the $d(d+1)/2$ -dimensional outer-product targets over-constrain the flight phase, causing coverage collapse (0.846–0.862), while the d -dimensional score vector attains comparable or better RMSE and stays valid. The $F^{-1/2}$ normalization is a wash versus raw outer products, confirming it is not worth its cost.

I ABLATION 5: ROBUSTNESS TO KERNEL CHOICE

Every kernel improves on the no-kernel baseline by $\approx 12\%$ (Table 8); the within-dataset spread across kernels ($\leq 3.4\%$) is of the order of Monte Carlo noise. The Linear kernel is rank-limited to $\sim$ feature-dimension modes, so it is adequate here but least future-proof for strongly non-linear populations; RBF (default) and Laplacian are the safe choices.

J ABLATION 6: INTERACTION WITH BLEND PARAMETER τ

Figure 5 and Table 9 show coverage does not collapse for any variant over $\tau \in [0, 1]$ (design-consistent estimators self-inflate as π peaks), so aggressive τ costs efficiency, not validity. Because NSK’s width is flat-to-decreasing in τ while baselines are U-shaped and inflate toward $\tau = 1$, NSK’s advantage rises from +3% to +45%; $\tau = 0.5$ is a safe near-optimal default for all methods.

K FULL BUDGET-SWEEP CI WIDTHS

Table 10 provides the numeric values underlying Figure 2.

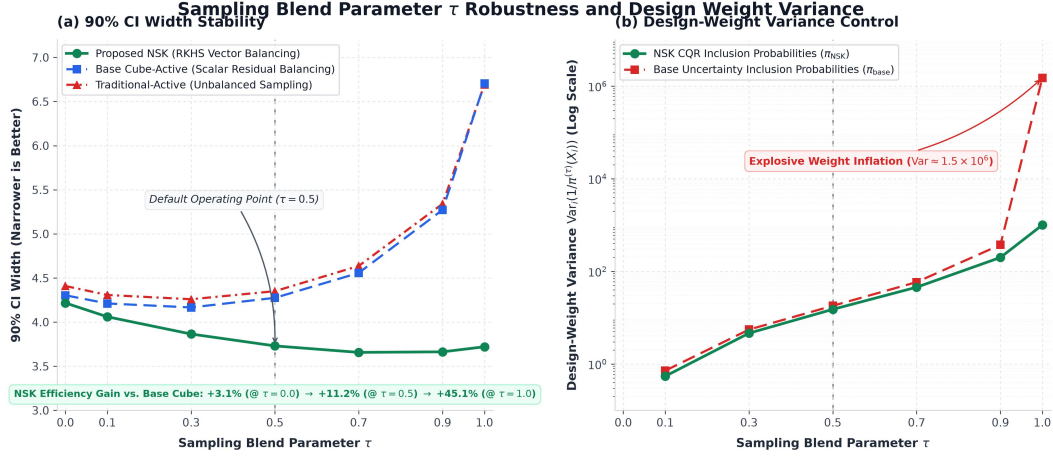


Figure 5: Blend parameter τ robustness on Bike Sharing. (a) 90% CI width across $\tau \in [0, 1]$: NSK stays tight while baselines inflate steeply near $\tau = 1$. (b) Design-weight variance $\text{Var}(1/\pi^{(\tau)})$ (log scale): NSK suppresses the explosive weight inflation ($\approx 1.5 \times 10^6$) that the base uncertainty target suffers.

Table 10: CI width across label budgets ($T = 500/\text{budget}$) for four representative datasets. **Bold** marks the narrowest per row; NSK is narrowest at every budget.

Dataset	Budget	Classical	Uniform	Traditional-Active	Base Cube-Active	NSK (Proposed)
Linear (synthetic)	0.05	0.4834	0.2816	0.2772	0.2291	0.2053
Linear (synthetic)	0.10	0.3328	0.1941	0.1908	0.1581	0.1439
Linear (synthetic)	0.15	0.2639	0.1538	0.1512	0.1262	0.1151
Linear (synthetic)	0.20	0.2217	0.1292	0.1270	0.1064	0.0968
Linear (synthetic)	0.25	0.1920	0.1120	0.1098	0.0926	0.0840
Nonlinear (synthetic)	0.05	1.0087	0.5258	0.4860	0.2955	0.2652
Nonlinear (synthetic)	0.10	0.6941	0.3617	0.3332	0.2040	0.1858
Nonlinear (synthetic)	0.15	0.5504	0.2866	0.2631	0.1629	0.1482
Nonlinear (synthetic)	0.20	0.4626	0.2409	0.2198	0.1377	0.1249
Nonlinear (synthetic)	0.25	0.4005	0.2087	0.1890	0.1197	0.1083
Bike Sharing	0.05	27.982	6.4176	6.3430	6.2128	5.3661
Bike Sharing	0.10	19.210	4.4061	4.3566	4.2638	3.7426
Bike Sharing	0.15	15.248	3.4953	3.4562	3.3949	2.9352
Bike Sharing	0.20	12.809	2.9412	2.8985	2.8541	2.4547
Bike Sharing	0.25	11.083	2.5480	2.5157	2.4857	2.0987
Superconductivity	0.05	4.7517	1.4781	1.4875	1.4716	1.2911
Superconductivity	0.10	3.2681	1.0230	1.0270	1.0105	0.8890
Superconductivity	0.15	2.5922	0.8113	0.8137	0.8028	0.7026
Superconductivity	0.20	2.1782	0.6816	0.6839	0.6744	0.5854
Superconductivity	0.25	1.8865	0.5904	0.5929	0.5849	0.5026